Predicting Secondary Market Price Trajectories of Physical Nintendo Games

CS 2756: Principles of Data Mining

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***Abstract*—I present a data mining pipeline that models and forecasts secondary market prices of physical Nintendo games from Game Boy Color through Nintendo 3DS. The system integrates four data sources (PriceCharting, IGDB, VGChartz, BLS CPI) to build a feature matrix covering 4,875 games and 4.6M+ monthly price records across seven discontinued consoles. Three LightGBM quantile regression models are trained on monthly price history and engineered features to produce price forecasts with per-game SHAP explanations. Backtesting against actual 2024–2025 prices yields a 1-year MAPE of 40.5% and a 2-year MAPE of 16.6%. An interactive HTML lookup tool displays both evaluation (Trained on data through 2023) and production (Trained on all data) forecasts through December 2030 for every game in the catalog.**

I. Introduction

The secondary market for retro physical video games has grown heavily over the past decade, largely driven by a shrinking supply and growing collector demand. Unlike stocks, there are no established tools for collectors to evaluate whether a game is fairly priced or where its value is heading.

This project aims to provide a solution to this problem by applying machine learning and data mining techniques. Given a game's current price and observable characteristics, where will its price be in 1–5 years? I focus on seven fully discontinued Nintendo platforms — Game Boy Color (GBC), Game Boy Advance (GBA), GameCube, Nintendo DS, Wii, Wii U, and Nintendo 3DS — including all physical releases from 1998 through the 3DS end-of-life in 2020. All systems are fully discontinued, meaning there is no need to account for mid-lifecycle volatility. The Nintendo Switch family is excluded as it remains an active platform.

To answer this, I built the following four components:

1) Data Pipeline: Fetched and/or scraped PriceCharting (catalog and price history), IGDB (game metadata), VGChartz (sales figures), and BLS CPI-U (inflation data) then filtered and merged into a single dataset of 4,875 games with 4.6M+ monthly price records. Hardware, Accessories, Titles with insufficient data were filtered out.
2) Feature Engineering: 50+ features were derived from the merged dataset to capture non-linear relationships between a game's market behavior and its observable characteristics. These include price velocity, volatility, and acceleration across multiple time windows, market and genre comparisons, franchise context, sales-based competitive positioning, and inflation-adjusted price anchors.
3) Forecasting Model: Three LightGBM quantile regression models ($\alpha \in \{0.10, 0.50, 0.90\}$) were trained to predict monthly log price changes. Forecasts are generated month-by-month through December 2030 with rolling feature updates at each step. Model performance was evaluated using SHAP explanations and backtesting against actual 2024–2025 prices.
4) Lookup Tool: A standalone HTML tool was built that displays per-game price forecasts and prediction intervals for every game in the catalog. The tool includes both an evaluation version (trained through 2023, showing backtest results) and a production version (trained on all available data, forecasting through December 2030).

II. Data Collection

A. Sources Overview

Four data sources were integrated to build the feature matrix. Table I summarizes each source, its collection method, and what it contributes.

TABLE I
Data Sources

Source	Method	Contribution
PriceCharting	JSON cursor API + HTML scraping	7,638-title catalog; 4.6M+ monthly price records (2006–2026)
IGDB API	REST API, fuzzy title matching	Genre, publisher, ESRB rating, franchise, critic/user scores, re-release flags
VGChartz	HTML scraping, fuzzy title matching	Lifetime units sold (NA and global, in millions)
BLS CPI-U	CSV download	Annual CPI-U averages for inflation adjustment

B. PriceCharting Catalog and Price History

The game catalog was built by reverse-engineering PriceCharting's hidden JSON API endpoint, found in the site's front-end JavaScript. By incrementing the cursor value on each request until the server returned an empty list, all 7,638 unique entries were collected across the seven target consoles. Each entry includes the game title, console, and a stable `pc_url` that is used as the join key throughout the rest of the pipeline.

Price history was collected by visiting each game's individual page and pulling the embedded `VGPC.chart_data` JavaScript object, which holds monthly price data (in cents) for up to six condition grades: *loose*, *CIB*, *new*, *graded*, *box only*, and *manual only*. Each scrape included a 1.5-second delay between requests and saved progress to a checkpoint file so the process could be restarted if interrupted. The final `price_history.csv` contains 4.6M+ rows going back to January 2006.

However, since Game Boy Color and Game Boy Advance games used cardboard packaging and were often thrown away, the market for those consoles primarily trades in loose cartridges. For this reason, *loose* pricing is used as the canonical condition for GBC and GBA, while *complete-in-box (CIB)* is used for GameCube, DS, Wii, Wii U, and 3DS. All price history, model training, and forecasts in this project are based on this canonical condition only.

C. IGDB Metadata

Game metadata was pulled from the IGDB v4 API. Since PriceCharting and IGDB use different naming conventions, titles were matched using fuzzy string matching (RapidFuzz). Fields retrieved include primary genre, publisher, ESRB rating, franchise membership, Metacritic critic and user scores, and re-release type (remake, remaster, port, or DLC).

D. VGChartz Sales Data

Lifetime sales figures (NA and global, in millions of units) were scraped from VGChartz and fuzzy-matched to the master catalog. Where NA figures were unavailable, global figures were used as a fallback.

E. Inflation Adjustment

BLS CPI-U annual averages were downloaded as a CSV to adjust for inflation. All prices throughout the pipeline are converted to constant 2025 dollars using

$$p_{\text{real}} = p_{\text{nominal}} \times \frac{\text{CPI}_{2025}}{\text{CPI}_t}, \quad \text{CPI}_{2025} = 322.153. \tag{1}$$

For years where CPI data was not yet available, the most recent year's value was carried forward.

F. Merging and Exclusions

The four sources were joined in two stages: the master game list (catalog + IGDB metadata) was merged with VGChartz sales on `igdb_id`, then merged with price features from `price_history.csv` on `pc_url`.

Games were removed from the dataset with the following rules:

1) Hardware and accessories: 57 regex patterns (controllers, chargers, handhelds) were used to identify and remove non-game products.
2) Sparse price history: games with fewer than 50 second-hand price records in their canonical condition were removed.

Variant editions (budget reprints, limited editions, retailer exclusives, bundles) were kept as separate rows with `is_variant=True` rather than being merged into the standard edition, since variant price premiums are themselves something the model should learn. After all exclusions, 4,875 games remained in the modeling dataset.

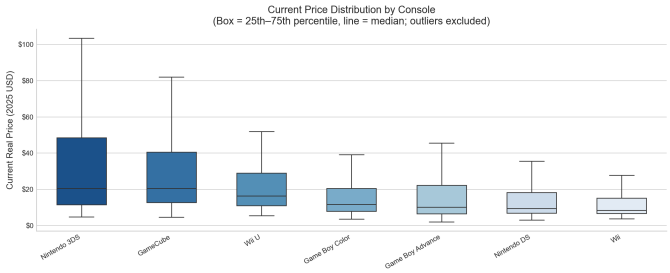

Fig. 1. Current real price distribution by console (box = 25th–75th percentile, line = median; outliers excluded).

III. Feature Engineering

All feature engineering is implemented in `features.py`. Features fall into six groups, all operating on CPI-deflated real prices in constant 2025 dollars.

A. Price Dynamics

This is the largest feature group and captures how a game's price has been moving relative to its own history. Five time windows are used: $w \in \{1, 3, 6, 24, 60\}$ months, plus a full-history window capped at 240 months. For each window, two values are computed:

Velocity ($\dot{p}_w$): a linear regression slope of real price over time within the window, expressed as annualized \$/yr.

Volatility (σ_w): the standard deviation of month-over-month absolute price changes within the window.

Additional features include:

- Absolute and percent price change over 1, 3, 6, and 12 months.
- Price acceleration: the difference between the velocity of the last 3 months and the 3 months before that, capturing whether a price trend is speeding up or slowing down.
- 52-week high/low ratios: current price divided by the trailing 12-month high and low.

B. Market Context

- Price vs. market median: for each year in a game's price history, the ratio of its real price to the median real price across the entire catalog that year is computed. The feature is the mean of these annual ratios, capturing

whether a title consistently trades above or below the broader market.

- Price vs. genre median: the current real price divided by the median real price of all games in the same genre.
- MSRP ratio: the real price at 60 months post-release divided by the inflation-adjusted launch MSRP for that console. Values above 1 mean the game has appreciated past its original retail price.
- Market liquidity score: the total number of monthly price records for a game, used as a measure of how actively it trades on the secondary market.

C. Temporal Features

- months_since_release: months from the game's release year to the reference year.
- months_since_discontinued: months from the console's discontinuation year to the reference year.
- real_price_at_12m, real_price_at_60m: the real price closest to 12 and 60 months post-release (within ± 3 months), anchoring the game's early price trajectory.

D. Franchise Features

For games belonging to a franchise in IGDB, four features describe the surrounding release context: total number of franchise entries in the catalog, days since the previous franchise release, days until the next one, and how many franchise entries released within ± 2 years of this game.

E. Competitive Positioning

Three features use VGChartz sales figures to place each game within its competitive context: log-transformed units sold, the best-selling title in the same genre and console within a ± 1 year release window, and this game's sales as a fraction of that benchmark. A cross-platform version captures the equivalent best-seller on other consoles in the same window.

F. Categorical and Metadata Features

Categorical inputs include console, genre, ESRB rating, is_variant, and variant_type. Publisher is encoded by replacing each publisher name with the average price change of that publisher's games in the training data, so the model can use it as a numeric input without seeing future prices. Re-release flags from IGDB (is_remake, is_remaster, is_port) are included as binary indicators, along with critic score, user score, and the score delta between them.

IV. Forecasting Model

A. Eval and Production Modes

The model runs in two modes. In eval mode, the price history is truncated at December 2023 and the model is trained only on data up to that point. Forecasts are then generated for January 2024 through December 2025 and compared against actual prices. In production mode, the model is trained on all available data and forecasts through December 2030. Both modes use the same architecture and feature set.

B. Target Variable

Rather than predicting the next absolute price directly, the model predicts the monthly price change:

$$y_t = \log\left(\frac{p_t}{p_{t-1}}\right) \tag{2}$$

where p_t is the real price in month t. This keeps the target scale-invariant across games ranging from \$2 to \$500+, and the inverse transform $p_t = p_{t-1} \cdot e^{y_t}$ always produces a positive price.

C. Model Architecture

Three LightGBM models are trained independently, one for each quantile: $\alpha \in \{0.10, 0.50, 0.90\}$. Together they produce a median forecast and a prediction interval for each game and month. All three share the same hyperparameters: learning rate 0.05, 63 leaves, minimum 20 samples per leaf, 80% feature and row subsampling, and early stopping after 50 rounds with no improvement on a 10% held-out validation set. The validation split is done at the game level meaning all rows for a given game go entirely into train or validation, never both.

D. Training Data

Two types of rows are used for training. Historical rows are built from the last 36 months of each game's real price history: each consecutive month-over-month step becomes one training row with the log-return as the target. Eval rows are the actual observed 2024–2025 prices treated as additional targets; these are included in production training so the model sees recent market behavior.

E. Sequential Inference

Forecasts are generated one month at a time, where each month's predicted price becomes the input for the next. Dynamic features like velocity and volatility are recomputed each step from the running prediction history so the model always sees an up-to-date price trend.

Market-wide and per-genre price trends are pushed forward using a linear regression fit to historical annual medians, and injected at each step so the model has current market context during inference.

F. Prediction Intervals

The interval width grows with time. At each step the single-step quantile spread is scaled by $\sqrt{t}$, where t is the number of months ahead:

$$\text{half-interval}_t = \frac{q_{90} - q_{10}}{2} \times \sqrt{t} \tag{3}$$

This prevents the lower bound from collapsing to zero and the upper bound from exploding in the future, while still reflecting increasing uncertainty over time. A confidence score is derived from the interval width relative to the predicted price.

G. Explainability

SHAP values are computed for the median model using a snapshot of each game at a 12-month forecast horizon, sampled to 2,000 games.

Figure 2 shows the top 20 features by mean absolute SHAP value. The two strongest predictors are the most recent known price prev_real_price and the current price level current_real_price, which together account for the majority of model impact.

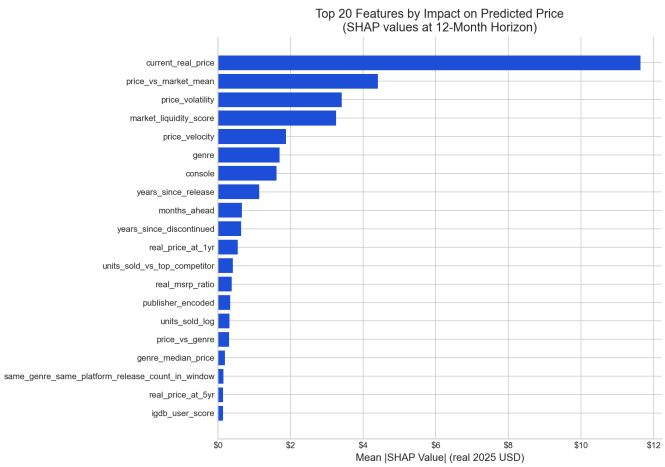

Fig. 2. Top 20 features by mean absolute SHAP value at the 12-month forecast horizon.

Beyond price level, the 24-month price velocity price_velocity_24m is the most influential trend feature, suggesting that medium-term momentum carries more signal than short-term fluctuations. How a game's price compares to the broader market price_vs_market_mean and its genre peers price_vs_genre also rank highly, indicating that relative market position matters. Genre and console both appear in the top 10, confirming that platform and game type affect average pricing patterns.

H. Evaluation

The eval model is assessed against actual 2024–2025 prices using RMSE, MAPE, R^2, and interval coverage. Results are split by horizon: months 1–12 (year 1) and months 13–24 (year 2). Table II summarizes the backtest results.

TABLE II
EVAL MODEL BACKTEST RESULTS

Metric	Year 1 (2024)	Year 2 (2025)
MAPE	40.5%	16.6%
RMSE	$59.23	$16.97
R^2	0.243	0.932
Interval Coverage (80%)	46.0%	79.5%
Games Evaluated	93.4%	35.7%

Year 2 covers only games with price data through 2025.

Year 1 performance is noticeably weaker, with a MAPE of 40.5% and an R^2 of 0.243. This is largely driven by cheaper games. As shown in Figure 4, games under $10 have significantly higher error than mid- and high-priced titles, and small absolute errors inflate percentage-based metrics when the denominator is near zero. The year 2 results (MAPE 16.6%, R^2 0.932) are stronger, but apply only to the 35.7% of games with price data extending through 2025, which skews toward more liquid and predictable titles.

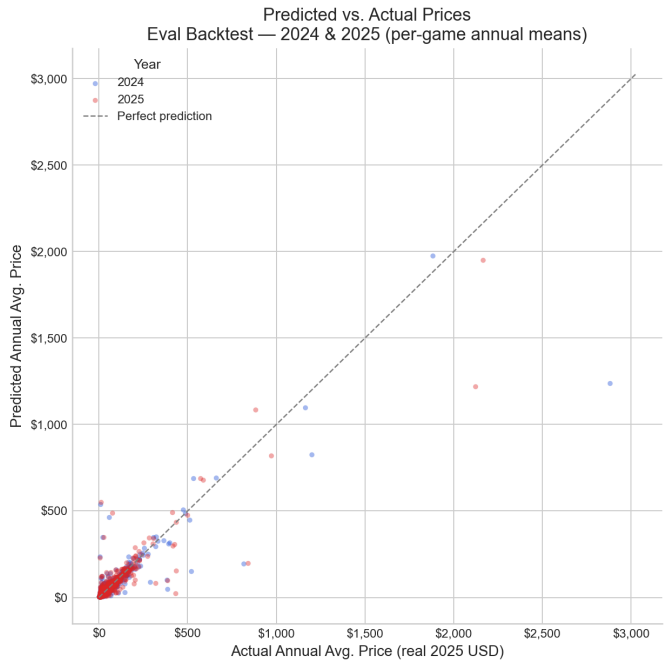

Fig. 3. Predicted vs. actual annual average prices for the eval backtest (2024 and 2025). Points along the dashed line indicate a perfect prediction.

Figure 3 shows the scatter of predicted vs. actual prices across both years.

Fig. 4. Mean and median MAPE by actual price tier.

V. LOOKUP TOOL

The final output of the pipeline is a standalone HTML dashboard built by build_lookup.py. The entire game catalog, price history, and forecasts are embedded directly in the HTML file as JSON. Two versions are generated: an eval version showing

the 2024–2025 backtest results alongside actual prices, and a production version showing forecasts through December 2030.

Each game's entry in the tool displays its console, condition grade (Loose or CIB), genre, publisher, ESRB rating, critic and user scores, NA units sold, and the number of price records it has. The chart toggles between monthly and annual granularity and between history only, forecast only, and both combined. Predictions are shown as a median line with a shaded Q10–Q90 interval that widens over time.

Figure **??** shows annual price history and production forecasts for the 12 games with the highest predicted 12-month price gain. Figure 6 shows the top 20 predicted gainers at the 12-month horizon ranked by absolute dollar increase. Figure 7 shows the December 2025 production forecast for the top 25 predicted gainers, with Q10–Q90 uncertainty bands.

VI. Limitations and Future Work

The model captures long-term pricing structure well but has several notable limitations. The most visible is accuracy on cheap games as shown in Figure 4, games priced under $10 have significantly higher MAPE than mid- and high-priced titles. This is partly a math problem: a $2 error on a $4 game is 50% MAPE, but in practice it is a negligible mistake. These games are also genuinely harder to model since small supply or demand shifts cause large percentage swings from a near-zero baseline.

Coverage is another constraint. Games that rarely trade on the secondary market may have months or years with no recorded price on PriceCharting, which limits what the model can learn about their trends and reduces forecast confidence. This is reflected in the backtest in which only 35.7% of the catalog had sufficient 2025 price data to evaluate at the 2-year horizon. The catalog itself is also a static snapshot from the time of scraping, meaning new price data and market shifts are not reflected until the pipeline is re-run.

An additional limitation is that the model has no awareness of current events. A viral video, a speedrunning tournament, or a nostalgia wave can move a game's price within days in ways that no historical feature can anticipate.

The most promising solution to this is integrating Google Trends search interest as a real-time demand signal. Preliminary testing showed that a rise in search interest for a game title typically precedes a rise in its secondary market price by 2–6 months, which would give the model early warning of demand shifts before they appear in price history. However, the Google Trends API imposes aggressive rate limiting that made it difficult to fetch data for all 4,875 games and fully integrate it within the time frame. As such, the integration remains a prototype and was excluded from the final model.

Beyond search trends, two other extensions are worth pursuing. First, an automated monthly refresh that re-scrapes PriceCharting, retrains the models, and republishes the lookup tool would keep forecasts current without manual effort. Second, inclusion of pre-GBC Nintendo consoles can be added to the current pipeline with minimal modification.

VII. Conclusion

This project shows that secondary market prices of physical Nintendo games can be modeled and forecasted using scraped price history, game metadata, and engineered features. The pipeline pulls from four data sources, builds a 4,875-game dataset, trains three LightGBM quantile regression models, and delivers the results through a standalone HTML lookup tool.

The backtest results show the model captures long-term pricing trends well, with a 2-year R^2 of 0.932 on liquid titles. It struggles more with cheap games where small absolute errors become large percentages. SHAP analysis confirms that the strongest predictors are the current price level and medium-term momentum, with genre and console adding meaningful signal on top.

The end result is a tool that gives collectors a data-driven starting point for evaluating whether a game is fairly priced and where its value is likely heading.

References

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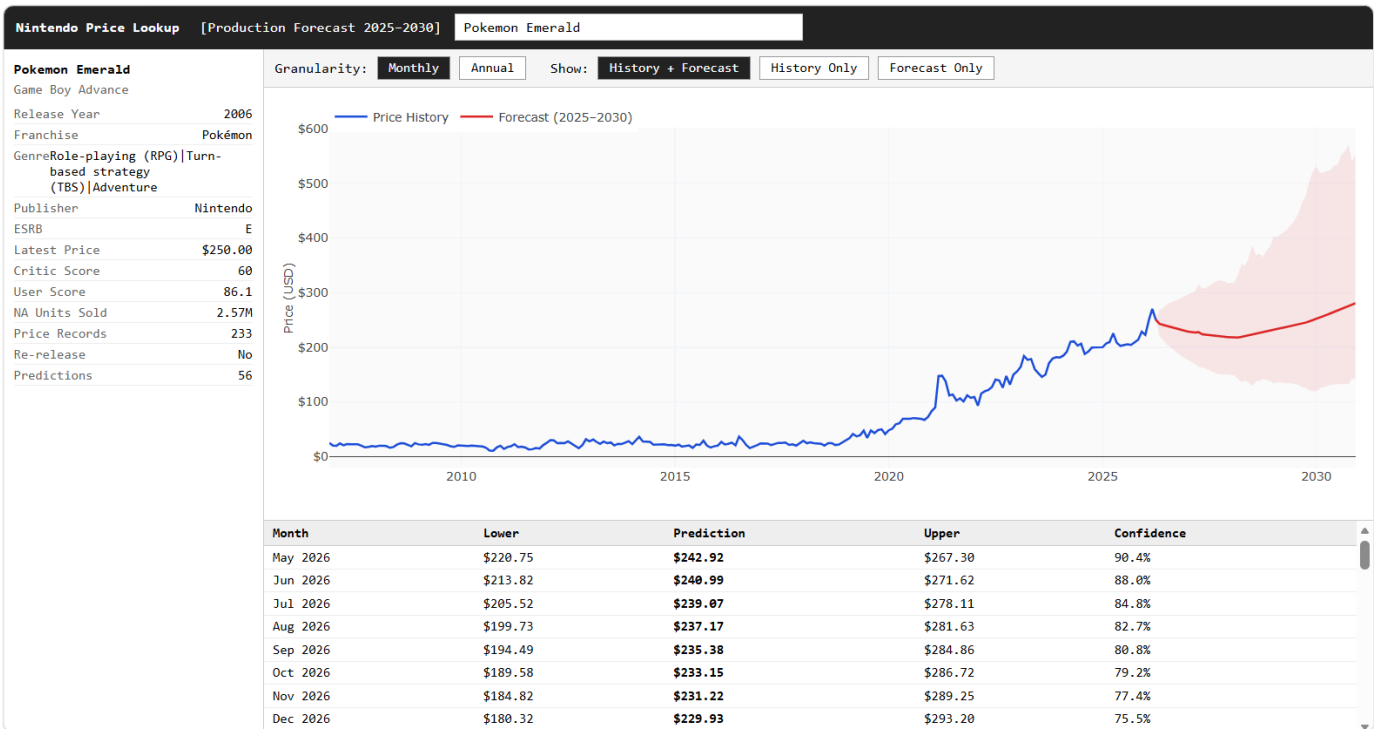
Appendix A
Project Repository

All source code and execution instructions can be found in the project repo:

https://github.com/gormley01/nintendo$_price_predictor$



Month	Lower	Prediction	Upper	Confidence
May 2026	$220.75	**$242.92**	$267.30	90.4%
Jun 2026	$213.82	**$240.99**	$271.62	88.0%
Jul 2026	$205.52	**$239.07**	$278.11	84.8%
Aug 2026	$199.73	**$237.17**	$281.63	82.7%
Sep 2026	$194.49	**$235.38**	$284.86	80.8%
Oct 2026	$189.58	**$233.15**	$286.72	79.2%
Nov 2026	$184.82	**$231.22**	$289.25	77.4%
Dec 2026	$180.32	**$229.93**	$293.20	75.5%

Fig. 5. The production lookup tool showing Pokemon Emerald (GBA).

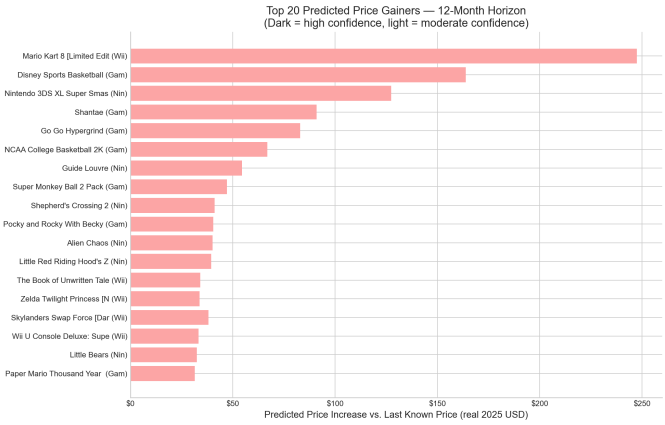


Fig. 6. Top 20 predicted price gainers at the 12-month horizon.

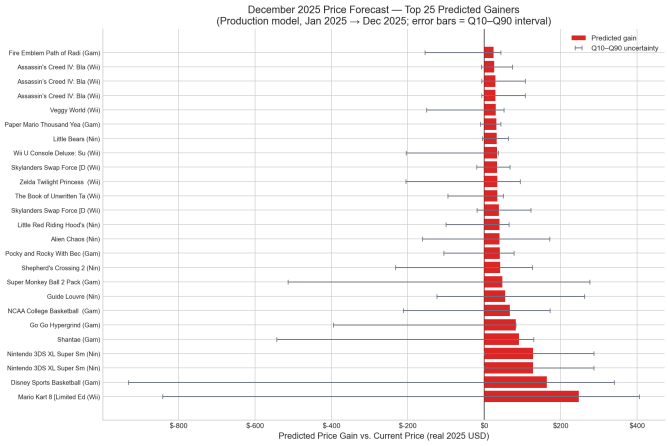


Fig. 7. December 2025 production forecast for the top 25 predicted gainers.